SAFETY DATA SHEET

This SDS complies with the requirements of JIS Z 7253:2019 (Japan)

Diethyl carbonate

1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Description:	**Diethyl carbonate**
Cat No. :	**114220000; 114220010; 114220025; 114220050; 114221000; 114225000; 114220100**
Synonyms	Ethyl carbonate
CAS No	105-58-8
Molecular Formula	C5 H10 O3

Details of the supplier of the safety data sheet

Supplier
UK entity/business name
Fisher Scientific UK
Bishop Meadow Road,
Loughborough, Leicestershire LE11 5RG, United Kingdom

Importer
Thermo Fisher Scientific Co., Ltd.
2-8 Shibaura 4-chome, Minato-ku, Tokyo
TEL: 0120-753-670 FAX: 0120-753-671

EU entity/business name
Thermo Fisher Scientific
Janssen Pharmaceuticalaan 3a, 2440 Geel, Belgium

Recommended use of the chemical and restrictions on use

Recommended Use Laboratory chemicals.

2. HAZARDS IDENTIFICATION

Flammable liquids	Category 3
Acute oral toxicity	Not classified
Acute dermal toxicity	Not classified
Acute Inhalation Toxicity	Not classified
Skin Corrosion/Irritation	Not classified
Serious Eye Damage/Eye Irritation	Not classified
Respiratory Sensitization	Not classified
Skin Sensitization	Not classified
Germ Cell Mutagenicity	Not classified
Carcinogenicity	Not classified
Reproductive Toxicity	Not classified
Specific target organ toxicity (single exposure)	Not classified
Specific target organ toxicity - (repeated exposure)	Not classified
Aspiration Toxicity	Not classified
Acute aquatic toxicity	Not classified for acute
Chronic aquatic toxicity	Not classified chronic
Hazardous to the ozone layer	Classification not possible

Label Elements

Signal Word
Warning

Hazard Statements
Flammable liquid and vapor

Precautionary Statements
Prevention
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking
Keep container tightly closed
Ground/bond container and receiving equipment
Use explosion-proof electrical/ventilating/lighting equipment
Wear protective gloves/protective clothing/eye protection/face protection
Use only non-sparking tools
Take action to prevent static discharges
Response
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish
Storage
Store in a well-ventilated place. Keep cool
Disposal
Dispose of contents/container to an approved waste disposal plant

Other Hazards
. This product does not contain any known or suspected endocrine disruptors.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Pure substance/mixture	Substance
Synonyms	Ethyl carbonate
Molecular Formula	C5 H10 O3
CAS No	105-58-8

Component	CAS No	Weight %	ENCS No.	ISHL No
Diethyl carbonate	105-58-8	>95	(2)-1699	(2)-1699

4. FIRST AID MEASURES

Inhalation
Remove to fresh air Get medical attention immediately if symptoms occur If not breathing, give artificial respiration

Eye Contact
Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes Get medical attention

Skin Contact
Wash off immediately with plenty of water for at least 15 minutes Get medical attention immediately if symptoms occur

Ingestion
Do NOT induce vomiting Get medical attention

Most important symptoms and effects
Difficulty in breathing Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

Self-Protection of the First Aider
Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination

Notes to Physician
Treat symptomatically

5. FIRE-FIGHTING MEASURES

Suitable Extinguishing Media
Water spray, carbon dioxide (CO2), dry chemical, alcohol-resistant foam Water mist may be used to cool closed containers
Extinguishing media which must not be used for safety reasons
No information available

Specific Hazards Arising from the Chemical
Flammable Risk of ignition Vapors may form explosive mixtures with air Vapors may travel to source of ignition and flash back Containers may explode when heated
Hazardous Combustion Products
Carbon monoxide (CO) Carbon dioxide (CO_2)

Flash Point	33 °C / 91.4 °F
Method -	No information available
Autoignition Temperature	445 °C / 833 °F
Explosion Limits	
Upper	No data available
Lower	No data available
Flammability (solid,gas)	Not applicable
Explosive Properties	explosive air/vapour mixtures possible
Oxidizing Properties	No information available
Sensitivity to Mechanical Impact	No information available
Sensitivity to Static Discharge	No information available

Protective Equipment and Precautions for Firefighters
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear

NFPA

Health	Flammability	Instability	Physical hazards
1	3	1	N/A

6. ACCIDENTAL RELEASE MEASURES

Personal Precautions
Use personal protective equipment as required Remove all sources of ignition Take precautionary measures against static discharges

Environmental Precautions
Should not be released into the environment See Section 12 for additional Ecological Information

Methods for Containment and Clean Up
Remove all sources of ignition Soak up with inert absorbent material Use spark-proof tools and explosion-proof equipment Keep in suitable, closed containers for disposal Take precautionary measures against static discharges

Refer to protective measures listed in Sections 8 and 13.

7. HANDLING AND STORAGE

Handling
Wear personal protective equipment/face protection Avoid contact with skin, eyes or clothing Avoid ingestion and inhalation Keep away from open flames, hot surfaces and sources of ignition Use only non-sparking tools Use spark-proof tools and explosion-proof equipment Take precautionary measures against static discharges

Hygiene Measures
Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

Storage
Keep containers tightly closed in a dry, cool and well-ventilated place. Flammables area. Keep away from heat, sparks and flame.

Specific Use(s)
Use in laboratories

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Exposure Guidelines
This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies

Biological occupational exposure limits
This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Exposure Controls

Engineering Measures
Ensure that eyewash stations and safety showers are close to the workstation location Ensure adequate ventilation, especially in confined areas Use explosion-proof electrical/ventilating/lighting equipment Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

Personal protective equipment

Eye Protection Wear safety glasses with side shields (or goggles) (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Natural rubber Nitrile rubber Neoprene PVC	See manufacturers recommendations	-	EN 374	(minimum requirement)

Inspect gloves before use.
Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)
Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g.

sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.
Remove gloves with care avoiding skin contamination.

Skin and body protection	Wear appropriate protective gloves and clothing to prevent skin exposure
Respiratory Protection	No protective equipment is needed under normal use conditions
Large scale/emergency use	Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Hygiene Measures	Handle in accordance with good industrial hygiene and safety practice
Environmental exposure controls	No information available

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	Colorless	
Physical State	Liquid	
Odor	Ether, sweet	
Odor Threshold	No data available	
pH	6.3	1 g/l aq.sol
Melting Point/Range	-43 °C / -45.4 °F	
Softening Point	No data available	
Boiling Point/Range	126 - 128 °C / 258.8 - 262.4 °F	
Flash Point	33 °C / 91.4 °F	**Method -** No information available
Evaporation Rate	No data available	
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	**Lower** 1.4 Vol%	
	Upper 11.0 Vol%	
Vapor Pressure	11 mbar @ 20 °C	
Vapor Density	4.07	(Air = 1.0)
Specific Gravity / Density	0.975	
Bulk Density	Not applicable	Liquid
Water Solubility	18.8 g/L	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	**log Pow**	
Diethyl carbonate	1.33	
Autoignition Temperature	445 °C / 833 °F	
Decomposition Temperature	>320 °C	
Viscosity	0.827 mPa.s at 20 °C	
Explosive Properties		explosive air/vapour mixtures possible
Oxidizing Properties	No information available	
Molecular Formula	C5 H10 O3	
Molecular Weight	118.13	

10. STABILITY AND REACTIVITY

Reactivity	None known, based on information available
Stability	Moisture sensitive
Hazardous Reactions	None under normal processing

Hazardous Polymerization Hazardous polymerization does not occur

Conditions to Avoid Incompatible products Excess heat Keep away from open flames, hot surfaces and sources of ignition Exposure to moist air or water

Incompatible Materials Strong oxidizing agents Strong bases Strong acids Reducing Agent

Hazardous Decomposition Products Carbon monoxide (CO) Carbon dioxide (CO_2)

Explosion Data

Sensitivity to Static Discharge No information available

Sensitivity to Mechanical Impact No information available

11. TOXICOLOGICAL INFORMATION

Product Information

Symptoms Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting
Ingestion Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea.
Inhalation May cause irritation of respiratory tract.
Skin May cause irritation.
Eyes May cause irritation.

Component Information

(a) acute toxicity;

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Diethyl carbonate	>15000 mg/kg (Rat)		>19.5 mg/L/4h (Rat)

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Based on available data, the classification criteria are not met

(d) respiratory or skin sensitization;
Respiratory Based on available data, the classification criteria are not met
Skin Based on available data, the classification criteria are not met

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

Other adverse effects .

12. ECOLOGICAL INFORMATION

Ecotoxicity effects Do not empty into drains

Component	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
Diethyl carbonate	Leuciscus idus: LC50 >500 mg/L/48H		EC50 >10000 mg/L/3H	

Persistence and Degradability Readily biodegradable
Persistence Soluble in water Persistence is unlikely based on information available.

Bioaccumulative Potential Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Diethyl carbonate	1.33	No data available

Mobility in soil The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

Hazardous to the ozone layer Based on available data, the classification criteria are not met

Ozone Depletion Potential This product does not contain any known or suspected substance

Persistent Organic Pollutant This product does not contain any known or suspected substance

Other adverse effects No information available

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

13. DISPOSAL CONSIDERATIONS

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

Other Information Waste codes should be assigned by the user based on the application for which the product was used Do not flush to sewer Can be landfilled or incinerated, when in compliance with local regulations

14. TRANSPORT INFORMATION

IMDG/IMO

UN-No UN2366
Proper Shipping Name DIETHYL CARBONATE
Hazard Class 3
Packing Group III

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code
Not applicable, packaged goods

Road and Rail Transport

UN-No	UN2366
Proper Shipping Name	DIETHYL CARBONATE
Hazard Class	3
Packing Group	III
Environmental hazards	No hazards identified

IATA

UN-No	UN2366
Proper Shipping Name	DIETHYL CARBONATE
Hazard Class	3
Packing Group	III

Domestic regulations	See section 15. If product is subject to the Fire Service Law, Poisonous and Deleterious Substance Control Law, High Pressure Gas Safety Law, Ship Safety Law, and/or the Civil Aeronautics Act, the requirements that are specific to each of the laws must be followed.
Special Precautions for User	No special precautions required

15. REGULATORY INFORMATION

National Regulations - Japan

Pollutant Release Transfer Register PRTR (2021 Amendment)
Not applicable

Industrial Safety and Health Law
Not applicable

Dangerous Substances
Industrial Safety and Health Law enforcement order Table 1 (related to article 6 and article 9-3)

Component	Dangerous Substances
Diethyl carbonate (CAS #: 105-58-8)	Flammable substance 4-3

Harmful Substances Whose Names Are to be Indicated on the Label
Not applicable

ISHL Notifiable Substances
Not applicable

Poisonous and Deleterious Substances Control Law
Not applicable

Component	CAS No	Poisonous and Deleterious Substances
Diethyl carbonate	105-58-8	Not applicable

Fire Service Law

Component	Fire Service Law
Diethyl carbonate	Group 4 (Flammable liquids) Hazard rank II Non-water soluble liquid

Ship (Marine Transportation) Safety Act
Flammable liquids - Ship (Marine Transportation) Safety Act and Ordinance Regarding Transport by Ship and Storage article 3 and Table 1

Component	Ships Safety Law
Diethyl carbonate (CAS #: 105-58-8)	Flammable liquids

Civil Aeronautics Act
Flammable liquids - Civil Aeronautics Act and Ordinance for Enforcement of the Civil Aeronautics Act article 194 and Table 1
See section 14 for more information

Port Regulation Law
Flammable liquids (Dangerous goods) - Act on Port Regulations article 21-2 and Enforcement Rules article 12 and Ministry of Transport Notification No. 547 of 1979 Appendix

International Regulations

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Diethyl carbonate	105-58-8	Not applicable	Not applicable	Not applicable	Not applicable

Component	CAS No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential
Diethyl carbonate	105-58-8	Not applicable	Not applicable	Not applicable

International Inventories
X = listed.

Component	ENCS	ISHL	EINECS	TSCA	IECSC	KECL	PICCS	AICS	DSL	NDSL
Diethyl carbonate	X	X	203-311-1	X	X	KE-04706	X	X	X	-

16. OTHER INFORMATION

Creation Date 19-Aug-2010
Revision Date 13-Dec-2024
Revision Summary Not applicable

Training Advice
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.
First aid for chemical exposure, including the use of eye wash and safety showers.

Key or legend to abbreviations and acronyms used in the safety data sheet

CAS - Chemical Abstracts Service
ISHL - Industrial Safety and Health Law (Japan)
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
AICS - Australian Inventory of Chemical Substances
NZIoC - New Zealand Inventory of Chemicals
CSCL - Chemical Substances Control Law Inventory
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
DNEL - Derived No Effect Level
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
NOEC - No Observed Effect Concentration
PBT - Persistent, Bioaccumulative, Toxic
TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
OECD - Organisation for Economic Co-operation and Development
BCF - Bioconcentration factor
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
ATE - Acute Toxicity Estimate
VOC - (Volatile Organic Compound)

Key literature references and sources for data
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

This SDS complies with the requirements of JIS Z 7252:2019 and JIS Z 7253:2019 (Japan)

Disclaimer
The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet